

Quantum Evolution and Clustering Probabilities in a Vortex Fluid: Analogy with a BEC

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Abstract—This note analyzes the unitary quantum dynamics of an N -vortex fluid governed by a pairwise logarithmic Hamiltonian. Starting from either a multi-branch cat state or a squeezed cluster state, we derive the probability that all vortices remain localized within a distance δ of a single vortex site after unitary evolution. We then apply a coordinate reflection operator and show that this probability remains unchanged due to the system's symmetry.

I. INTRODUCTION

Point vortices in two-dimensional ideal fluids exemplify Hamiltonian many-body interactions and chaos [3]. Their quantum version features nontrivial spreading and entanglement and an analogy with a BEC with N atoms can be drawn. We study the quantum Hamiltonian for N point vortices with circulations Γ_α and complex positions $z_\alpha = x_\alpha + iy_\alpha$:

$$H = -\frac{1}{4\pi} \sum_{\alpha < \beta} \Gamma_\alpha \Gamma_\beta \log |z_\alpha - z_\beta|.$$

Classically, the coordinates are cross-coupled:

$$\dot{x}_\alpha = \frac{1}{\Gamma_\alpha} \frac{\partial H}{\partial y_\alpha}, \quad \dot{y}_\alpha = -\frac{1}{\Gamma_\alpha} \frac{\partial H}{\partial x_\alpha}.$$

II. QUANTUM INITIAL STATES AND EVOLUTION

Quantization promotes $\{x_\alpha, y_\alpha\}$ to operators with canonical commutators. The unitary evolution is

$$|\psi(T)\rangle = e^{-iHT/\hbar} |\psi(0)\rangle.$$

We consider two initial states:

- **Cat state:** an equal superposition of clusters with all vortices coinciding at each of the N sites.
- **Squeezed state:** a single branch with relative coordinates squeezed along a chosen quadrature.

III. CLUSTER PROBABILITY

We define the barycenter and relative coordinates:

$$Z_B = \frac{1}{N} \sum_{\alpha} z_\alpha, \quad \xi_j = z_j - Z_B.$$

A measurement checks if all ξ_j satisfy $|\xi_j| < \delta$. Due to the log repulsion, the relative coordinates spread with time:

$$P(\delta, T) \approx \frac{\delta^4}{[\sigma_0^2 + V^2 T^2]^2}, \quad (N = 3),$$

where σ_0 is the initial cluster width and V characterizes the spreading rate.

IV. REFLECTION OPERATOR

We define a reflection operator:

$$R : z_\alpha \mapsto 2z_\alpha^{(0)} - z_\alpha.$$

This maps $\xi_j \mapsto -\xi_j$ but keeps Z_B unchanged. Since the Hamiltonian depends only on pairwise distances, it is invariant under R . The measurement region $|\xi_j| < \delta$ is also invariant. Therefore,

$$P_R(\delta, T) = P(\delta, T) \quad \text{for both cat and squeezed states.}$$

V. CONCLUSION

The probability that all vortices remain within a small cluster decays polynomially in time due to the logarithmic repulsion. This can be thought of as the reverse of BEC formation. Applying a reflection operator has no effect on this probability due to the symmetry of the Hamiltonian and measurement condition. For squeezed states, the spreading rate may differ, but the invariance argument holds.

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